Google Play Store Analytics

App Performance, Ratings, Installs, Reviews & Sentiment Analysis

Project Report

Tools: Power BI • DAX • Power Query • Data Modeling

Prepared as a Data Analytics Portfolio Project

1. Executive Summary

This project analyzes Google Play Store applications to understand app performance, user engagement, ratings, installs, pricing, content ratings, update activity, and review sentiment. An interactive Power BI solution was developed to convert the raw application and review data into business-focused insights.

The analysis shows that the dataset is strongly dominated by free and relatively small applications. Game and Communication are among the leading categories by install volume, while Events records the highest average rating among the analyzed categories. Review sentiment also shows a clear relationship with rating groups: high-rated apps contain substantially more positive sentiment, whereas low-rated apps show more negative sentiment.

The final solution consists of a structured data model, DAX measures, business-question analysis, and a three-page Power BI dashboard designed for interactive exploration.

2. Business Problem

The Google Play Store contains a large and diverse ecosystem of applications. For analysts, developers, and business stakeholders, simply knowing the number of apps is not enough. They need to understand which categories attract users, how ratings and reviews relate to popularity, whether pricing is associated with satisfaction, how frequently applications are updated, and what users express in their reviews.

The project therefore focuses on turning application-level data into actionable insights that can support app-market evaluation, performance benchmarking, and user-experience analysis.

3. Project Objectives

- Analyze the overall structure and characteristics of Google Play Store applications.
- Compare free and paid applications across key performance and engagement metrics.
- Identify high-performing categories, genres, and applications based on installs, reviews, and ratings.
- Examine relationships between installs, app size, price, and ratings.
- Analyze application update activity over time.
- Evaluate review sentiment and compare sentiment patterns across rating groups and categories.
- Build an interactive Power BI dashboard for clear and decision-oriented reporting.

4. Dataset Overview

The project uses Google Play Store application data together with review-level sentiment information. The application data contains fields describing app identity, category, rating, reviews, installs, size, price, content rating, genre, type, and last update information. The review data provides textual-review-derived sentiment information including sentiment category and sentiment polarity.

Analytical Area	Examples of Fields / Metrics
App information	App name, category, genre
Engagement	Reviews, installs
Quality	Rating
Commercial model	Type, price
Technical characteristics	Size, last updated
Audience	Content rating
Reviews	Sentiment, sentiment polarity

5. Data Preparation & Transformation

Data preparation was performed to make the source data suitable for analysis and Power BI reporting. Key preparation activities included standardizing fields, handling missing or unusable values, converting Last Updated into a date/datetime field, and preparing numerical install values for aggregation and binning.

Install values were transformed into numeric form so that categories and applications could be compared using total installs. Install-range bins were created for grouped analysis of average ratings. Review sentiment data was prepared using sentiment categories and a polarity score ranging from -1 to +1.

The review analysis also required attention to missing sentiment values. Unrated applications were kept separate when interpreting the relationship between rating and sentiment so that missing ratings were not treated as meaningful rating groups.

6. Data Modeling

The Power BI model uses a fact-and-dimension structure to improve filtering, aggregation, and analytical consistency. The model includes application dimensions and a review fact table, with relationships established through application keys.

A dedicated measures table was used to organize analytical calculations. This approach separates reusable business measures from source columns and makes the dashboard easier to maintain.

Key analytical measures include average rating, review counts, install totals, average sentiment polarity, and grouped sentiment metrics.

7. Analytical Approach

7.1 Descriptive Analysis

Descriptive analysis was used to understand the distribution of applications by category, size, type, content rating, and update year. KPI cards provide high-level measures while charts provide category-level and application-level comparisons.

7.2 Relationship Analysis

Scatter plots and trend lines were used to investigate relationships such as installs versus ratings, app size versus installs, and price versus ratings. These relationships were interpreted cautiously because visible trends were generally weak and did not imply causation.

7.3 Binned Analysis

Install ranges were created to compare average ratings across different levels of app adoption. This made it possible to identify whether highly installed applications systematically received higher or lower ratings.

7.4 Sentiment Analysis

Review sentiment was analyzed using positive, neutral, and negative classifications together with sentiment polarity. High-, medium-, and low-rated application groups were compared to identify whether user sentiment aligned with app ratings.

8. Business Questions & Key Findings

8.1 Basic-Level Findings

Average rating: The average app rating is 4.17/5, indicating generally positive ratings across the dataset.

App size: Apps below 10 MB form the largest size group at approximately 5.0K apps.

Free vs paid: Approximately 92.17% of apps are free and 7.83% are paid.

Content rating: Everyone is the dominant content-rating group, with approximately 7.9K apps.

Top installs: Several major applications reach approximately 1B installs, including major Google services and large social platforms.

Rating 4+: 6,286 apps have a rating of 4.0 or higher.

Reviews by type: Free apps average approximately 234.27K reviews versus 8.72K for paid apps.

Average size by category: Game has the highest average app size at approximately 42 MB.

Updates in 2018: 6,284 apps were updated during 2018.

8.2 Medium-Level Findings

Installs vs ratings: The scatter plot shows a slight positive relationship, but the variation indicates installs alone do not strongly determine ratings.

Highest-rated categories: Events leads with an average rating of 4.44, followed by Education and Art & Design at 4.36.

Price vs ratings: Price and rating show a slight negative relationship; higher pricing does not translate into higher ratings.

Content rating vs average rating: Average ratings are relatively consistent across content-rating groups, approximately 4.1–4.3.

Genres above 1M installs: Tools leads with 172 apps above 1M installs, followed by Action (128) and Photography (123).

Update trend: Update activity increased substantially over time and peaked at approximately 6.3K apps in 2018.

App size vs installs: The relationship is slightly positive, but app size alone is not a strong predictor of popularity.

Highest review counts: Facebook has approximately 78.2M reviews, followed by Instagram at 66.6M and Messenger at 56.6M.

Free vs paid by content rating: Free apps dominate every content-rating group; overall there are 8,902 free apps versus 756 paid apps.

Top categories by installs: Game leads with 13.9B installs, followed by Communication (11.0B), Tools (8.0B), Productivity (5.8B), and Social (5.5B).

8.3 Advanced-Level Findings

Update trend and seasonality: Yearly update activity rose sharply and peaked in 2018. Because the data is aggregated by year, it does not support a reliable monthly seasonal conclusion.

Rating by install range: Average ratings range from approximately 4.04 to 4.37. The 100M+ group has the highest average rating at approximately 4.37, while the 10K–100K group has the lowest at approximately 4.04.

Review sentiment: Positive sentiment is 38.08% among high-rated apps versus 13.89% among low-rated apps. Negative sentiment is 12.56% for high-rated apps versus 18.61% for low-rated apps.

Sentiment polarity: Average polarity is 0.19 for high-rated apps, 0.11 for medium-rated apps, and -0.07 for low-rated apps.

Genre and ratings: Ratings vary by genre; examples in the analysis range from 3.90 to 4.60, showing that some genre combinations perform better than others.

Advanced Q1 note: The earlier visual for the free/paid content-rating question did not correctly represent the required dimensions. A final report should only include that question after the corrected visual is verified.

9. Dashboard Development

The final Power BI solution contains three dashboard pages. The pages were designed to move from high-level overview to deeper app-performance and review-sentiment analysis.

Page 1 — Executive Overview

The overview page provides KPI cards and summary visuals for the overall application ecosystem, including app counts, ratings, installs, reviews, app type, content rating, and other high-level performance indicators.

Page 2 — App Performance & Engagement

The second page focuses on application performance and engagement, including category and genre comparisons, install ranges, review volume, app ratings, and relationships between performance metrics.

Page 3 — Reviews & Sentiment Analysis

The third page focuses on review sentiment and rating groups. It includes sentiment distribution, sentiment by rating group, average sentiment by category, and sentiment polarity analysis.

10. Key Business Insights

• **Free applications dominate the ecosystem.** More than nine out of ten applications in the analyzed dataset are free.

• **Small apps are common.** The <10 MB size group contains the largest number of applications.

• **Game and Communication lead adoption.** These categories have the largest install totals among the analyzed categories.

• **Popularity and satisfaction are different dimensions.** Apps with very large review volumes are not necessarily the highest-rated applications.

• **Price is not a strong indicator of quality.** The price-rating relationship is weak and slightly negative.

• **Updates increased substantially.** Application update activity grew strongly toward 2018.

• **User sentiment aligns with ratings.** High-rated applications show more positive sentiment, while low-rated applications show more negative sentiment.

11. Business Recommendations

• Prioritize user experience and review quality rather than relying only on install growth as a success metric.

• For paid apps, benchmark pricing against comparable applications because higher prices do not automatically produce higher ratings.

• Monitor negative review themes in low-rated applications to identify recurring user-experience problems.

• Use category and genre benchmarks to identify segments with stronger ratings and adoption.

• Maintain regular application updates, while using user feedback to prioritize meaningful improvements.

• For highly installed apps, track rating and sentiment together to ensure scale is not achieved at the expense of user satisfaction.

12. Limitations

• The dataset represents a specific snapshot of the Google Play Store and may not reflect the current market.

• Install values are grouped/rounded in the source data, so exact install totals should be interpreted as reported estimates.

• Yearly update data does not provide sufficient detail for reliable monthly or seasonal analysis.

• Sentiment analysis depends on available review text and sentiment classification; missing sentiment values exist.

• Observed relationships such as installs versus ratings are associations and should not be interpreted as causal effects.

• One advanced business question required visual correction before a defensible conclusion could be reported.

13. Conclusion

The Google Play Store Analytics project demonstrates an end-to-end Power BI analytics workflow: data preparation, dimensional modeling, DAX measure development, exploratory analysis, relationship analysis, binned analysis, sentiment analysis, and interactive dashboard design.

The analysis shows a market dominated by free applications, strong adoption in categories such as Game and Communication, generally positive ratings, and a meaningful connection between review sentiment and rating groups. The resulting dashboard provides a practical framework for exploring app performance, engagement, and user satisfaction.

14. Technology Stack

Tool	Purpose
Power BI	Data modeling, DAX measures, interactive dashboards and visualization
Power Query	Data cleaning and transformation
DAX	KPI and analytical measure development
Data Modeling	Fact/dimension structure and relationship-based analysis
Markdown	Project documentation and data dictionary

15. Supporting Documentation

The project is supported by a Data Dictionary describing the dataset fields and a Question-Wise Analysis document containing detailed findings and insights for the business questions. These documents complement the Power BI dashboard and provide additional context for review and presentation.